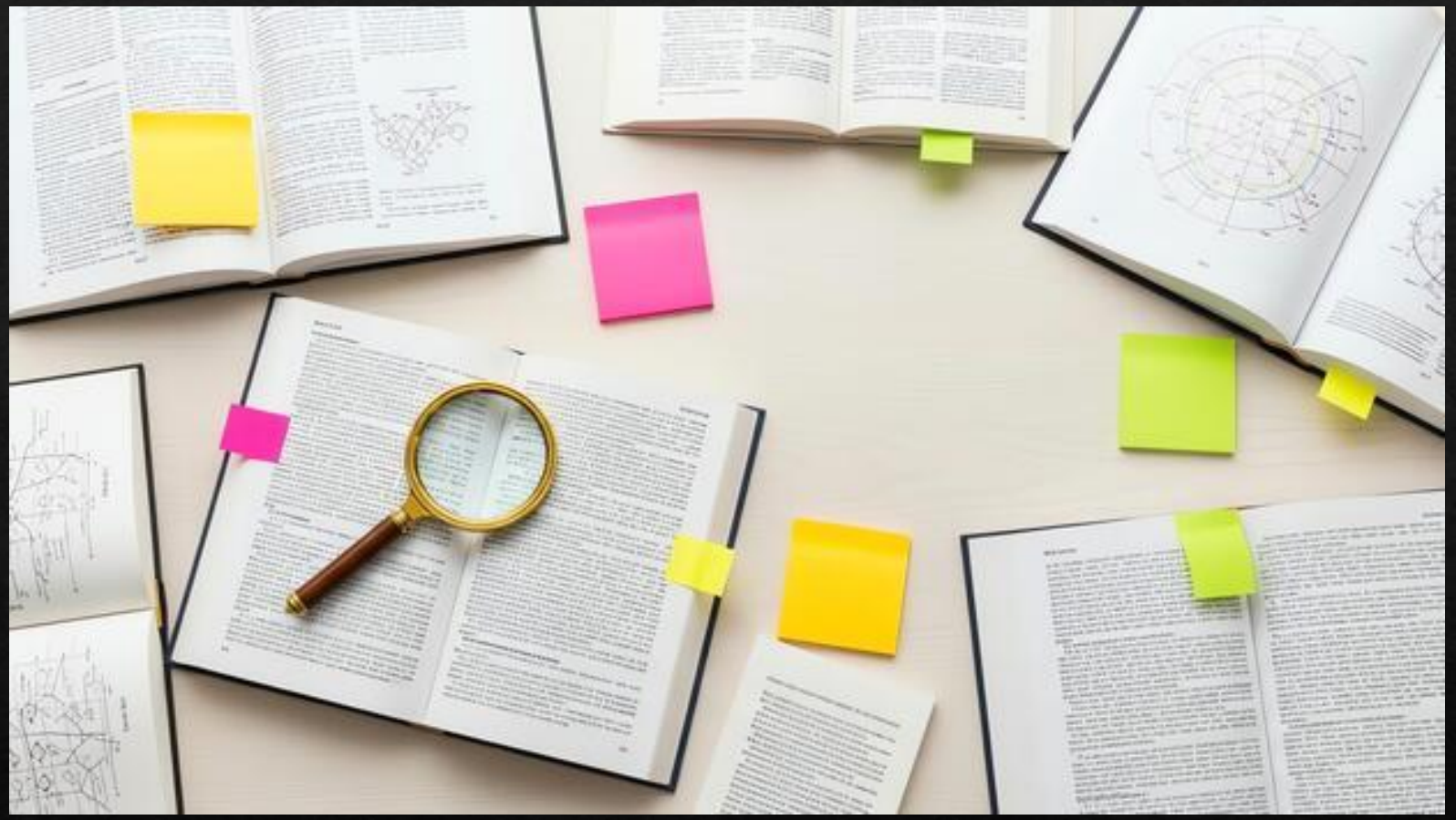
Indexing



Search a topic without index

Search a topic with index

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Imagine karo apke paas 10 lakh users ka data hai

aur aap query chalate ho:

```
db.users.find({ email: "abc@gmail.com" })
```

Agar indexing nahi hai… to MongoDB pura database scan karega

But agar indexing hai… to direct result milta hai in milliseconds

What is Indexing?

Simple language me Index ek shortcut / reference hota hai Jaise book me index hota hai (page number find karne ke liye)

Without Index: Full Collection Scan (slow ❌)

With Index: Direct search (fast ☑)

How Index Works Internally

MongoDB uses B-Tree structure

Data sorted form me store hota hai

Example: db.users.createIndex({ email: 1 })

1 = ascending

-1 = descending

Types of Indexes (Core Part)

1. Single Field Index
2. Compound Index
3. Multikey Index
4. Text Index

1. Single Field Index

Ek field pe index

```
db.users.createIndex({ email: 1 })
```

Use case:
Search by one field

2. Compound Index

Multiple fields pe index

```
db.users.createIndex({ name: 1, age: -1 })
```

Important:

- ❑ Order matters

Works for:

- ❑ Name
- ❑ name + age
- ❑ ❌ Not for only age

3. Multikey Index

Created on an array field

```
{ name: "Manas", skills: ["JS", "MongoDB"] }
```

Example data:

```
db.students.createIndex({ skills: 1 })
```

MongoDB automatically multiple entries banata hai

4. Text Index

Full-text search ke liye

```
db.posts.createIndex({ content: "text" })
```

Search:

```
db.posts.find({ $text: { $search: "mongodb tutorial" } })
```

Useful for:

- ❑ Blogs
- ❑ Search features

Features:

1. Tokenization
   Sentence → words me tod diya jata hai
   "Mongo is powerful" → mongo, powerful

2. Stop Words ignore hote hain
   Words jaise "is", "the", "a" ignore ho jate hain

3. Stemming
   "running", "runs" → "run" treat hoga

4. Score (Relevance)

Most relevant result pehle aayega

```
db.articles.find(
  { $text: { $search: "MongoDB" } },
  { score: { $meta: "textScore" } }
).sort({ score: { $meta: "textScore" } })
```

5. Multiple fields pe text index

```
db.articles.createIndex({
  title: "text",
  content: "text"
})
```

Dono fields me search hoga

Note: Text Score Factors:

- ❑ Term Frequency (TF)Word kitni baar aaya hai
- ❑ Field Length (Normalization)
  - Chhota field (title) → zyada impact
  - Bada field (content) → score dilute
- ❑ Match Quality
  - Exact word match vs weak match
- ❑ Number of matched terms
  - Kitne search words match hue

Important Limitations

1. Ek collection me sirf 1 text index
   But multiple fields include kar sakte ho

2. Exact phrase ke liye quotes

```
$search: "\"MongoDB index\""
```

3. Regex jaisa flexible nahi
   Ye full-text search hai, pattern matching nahi

More About Indexing

unique: true

Ensures duplicate values allow nahi honge

```
db.users.createIndex({ email: 1 }, { unique: true })
```

Use case: Email, Username, Phone number

Important:

- ❑ Agar collection me already duplicate hai → index create fail ho jayega
- ❑ Null bhi ek baar hi allowed hota hai (unless sparse/partial use karo)

partialFilterExpression

Index sirf selected documents par banta hai (full collection pe nahi)

```
db.users.createIndex(
  { age: 1 },
  { partialFilterExpression: { age: { $gt: 18 } } }
)
```

Sirf age > 18 wale docs indexed honge

Use case: Active users only, Paid users only, Filtered optimization

Benefit:

- ❑ Smaller index
- ❑ Faster writes (kyunki sab docs index nahi ho rahe)

expireAfterSeconds (TTL Index)

Auto delete documents after some time ⏳

```
db.sessions.createIndex(
  { createdAt: 1 },
  { expireAfterSeconds: 3600 }
)
```

Use case: OTP, Sessions, Logs

Notes:

- ❑ Background process run hota hai (instant delete nahi hota)
- ❑ Sirf date field pe kaam karta hai

4. Covered Query

- ❑ Jab query ka saara data index se hi mil jaye
- ❑ DB ko actual document fetch hi nahi karna padta

```
db.users.createIndex({ name: 1, age: 1 })

db.users.find(
  { name: "Manas" },
  { name: 1, age: 1, _id: 0 }
)
```

Benefit: Super fast , Disk read kam

Covered nahi hoga agar:

- ❑ extra field mang li
- ❑ _id include ho gaya (by default hota hai)

Winning Plan (Query Planner)

- MongoDB decide karta hai kaunsa index use kare
- In case of multiple index on same field, MongoDB tests multiple indexes once, picks the fastest (winning plan), and caches it for future similar queries to avoid re-evaluation.

Index Cardinality

Unique values kitne hain field me. High cardinality fields are best for indexing

Example:

- gender → low cardinality
- email → high cardinality

Sparse Index

- Sirf un docs pe index banega jisme field exist karta hai
- Missing fields ignore honge

```
db.users.createIndex({ phone: 1 }, { sparse: true })
```

Index Performance Optimization

Check Index Usage

```
db.users.find({ email: "abc@gmail.com" }).explain("executionStats")
```

Important fields:

COLLSCAN ❌ (bad)

IXSCAN ☑ (good)

Tips:

Always index frequently queried fields

Use compound index smartly

Avoid unnecessary indexes

When to Use Indexes

- Large data (1000+ docs)
- Frequently searched fields
- Sorting and filtering

Note:

Companies always use indexing for:

- Search
- Filtering
- Recommendations

When NOT to Use Indexes

- When collection is small
- When Fields with low variation (like gender)
- When queries are complex (multiple fields)
- When the collection is frequently updated (more write operations)

Note:

Indexing slows write process.
indexing slows down: insert, update, delete

"Indexing is a trade-off:
Read fast, Write slow."

Thank You